

# Dana Ferranti

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## CONTACT INFORMATION

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6823 St. Charles Avenue  
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✉ **E-mail**  
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## RESEARCH INTERESTS

- Computational methods for low Reynolds number fluid dynamics.
- Emergence of synchronization in fluid-solid body interactions.

## EDUCATION

**Tulane University**, New Orleans, LA 2017–2023 (expected)

- PhD in Mathematics department.
- Advisor: Dr. Ricardo Cortez.

**Clark University**, Worcester, MA. 2010–2014

- BA, Mathematics and computer science.

## RESEARCH EXPERIENCE

- **Tulane University** 2017-present  
Center for Computational Science in Mathematics Department.
  - Extending the method of regularized stokeslets by using exact integration over triangulated surfaces. Using minimal models of interacting bodies in viscous flows to study the dynamical differences that arise in models that incorporate weak inertia versus those that assume the fluid inertia is negligible.
- **Massachusetts General Hospital** 2016–2017  
Physics Research in Department of Radiation Oncology.
  - Using theoretical models to demonstrate the value of prior knowledge in determining causal relationships in complex networks, with applications to machine learning in medicine.
  - Advisor: Dr. David Craft.

## TEACHING EXPERIENCE

**As instructor**

- Probability & Statistics I (Math 1110). Spring 2023  
*Elementary probability theory and statistics*
- Introduction to Applied Math (Math 2240). Fall 2021  
*Ordinary differential equations for engineers/physicists*

**As teaching assistant**

- Introduction to Applied Math (Math 2240). 2019, 2020, 2021
- Linear algebra (Math 3090). 2020
- Calculus I (Math 1210). 2017, 2019
- Calculus II (Math 1220). 2018, 2020
- Calculus III (Math 2210). 2018

## SERVICE AND OUTREACH

- President of AMS Graduate Student Chapter 2019-2021
- Mathematics department tea time organizer 2018-2022
- Treasurer of AMS Graduate Student Chapter 2017-2019
- Member of Inclusivity in Mathematics Task Force at Tulane (IMTF) 2020-present

## TALKS

- *An Extension to the Method of Regularized Stokeslets*  
Special session on Recent Developments in Numerical Methods for PDEs, Joint Math Meetings 2023 (January 4, 2023)
- *Computational Modeling of Bodies Immersed in Viscous Fluids*  
Hunter College Applied Math Seminar (November 3, 2022)

## CONFERENCES

- Joint Math Meetings in Boston, MA (January 2023)
- SIAM Annual Meetings in Pittsburgh, PA (July 2022)
- Blackwell-Tapia Conference at IMSI in Chicago, IL (Nov 2021)
- Math for All in New Orleans (2020 and 2021)
- Scientific Computing Around Louisiana (2018, 2019, 2020)

## REFERENCES

**Dr. Ricardo Cortez**  
Tulane University